ARCHITECTURAL SPECIFICATION

Amagi-TRACE v1.0

Bidirectional Hazard-to-Verification Traceability

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Bidirectional traceability matrix per IEC 61508-1 §7.9, ISO 26262-4 §7.4.1, and DO-254 §6.2 linking hazards (H-01–H-08), safety requirements (SR-01–SR-16), invariants (INV-001–006, defined in NIPU v1.1), hardware components, formal verification properties, and test cases. Forward and backward coverage verified; four gaps identified with remediation plan. Aligned with: Framework v2.0, NIPU v1.1, MEM v1.1, HARA v1.0, FMEDA v1.0, Master Parameter Sheet v2.0, S-APL v2.0, Charter v1.2.

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Section 1: Executive Summary

This Traceability Matrix establishes the bidirectional traceability required by IEC 61508-1 §7.9, ISO 26262-4 §7.4.1, and DO-254 §6.2 between the following six categories of artefacts: (a) hazards identified in the AMAGI HARA v1.0, designated H-01 through H-08; (b) safety requirements derived from those hazards, designated SR-01 through SR-16; (c) AMAGI invariants that implement the safety requirements, designated INV-001 through INV-006; (d) hardware components that realize the invariants within the NIPU and MEM clusters; (e) formal verification properties that prove compliance with the invariant definitions; and (f) test cases that validate the implementation against the verification properties. The matrix is structured to support both forward and backward traceability, ensuring that no hazard is left without mitigation and that no verification activity exists without a clear link to a safety requirement.

The matrix demonstrates complete coverage in the following sense: every hazard identified in the HARA traces forward through at least one safety requirement to at least one invariant, and every invariant traces forward to at least one verification property. Conversely, every verification property traces backward through at least one invariant to at least one safety requirement, and every safety requirement traces backward to at least one hazard. This bidirectional chain of evidence is the fundamental requirement for certification under IEC 61508, ISO 26262, and DO-254, and its existence is a precondition for any independent assessment of the AMAGI safety case. The matrix identifies four gaps where verification evidence is incomplete, and provides a remediation plan for each gap with clear milestones and responsibilities.

The significance of this document extends beyond regulatory compliance. The traceability matrix serves as the primary tool for impact analysis when requirements change: if a hazard is reclassified or a new hazard is identified, the matrix immediately reveals which safety requirements, invariants, components, and verification properties are affected. Similarly, if a verification property fails during testing, the matrix reveals which hazards and safety requirements are at risk. This ability to trace the ripple effects of any change through the entire safety argument is essential for maintaining the integrity of the AMAGI safety case over the full lifecycle of the product.

Section 2: Purpose and Scope

The traceability matrix serves three distinct purposes within the AMAGI safety lifecycle. First, it demonstrates to certification auditors that no hazard identified in the HARA is left without at least one mitigation path that can be traced through safety requirements, invariants, components, and verification evidence. This completeness argument is the core of the IEC 61508 requirement for systematic capability, and it is the primary artifact that an assessor will examine when evaluating whether the AMAGI safety argument is well-founded. Without a complete traceability matrix, an assessor cannot determine whether the set of invariants is sufficient to address all identified hazards, and the safety case collapses into an unstructured collection of claims without evidence of completeness.

Second, the traceability matrix identifies gaps where an invariant lacks sufficient verification evidence. A gap exists when a safety requirement has been derived from a hazard and an invariant has been defined to implement that requirement, but no formal verification property or test case exists to confirm that the invariant holds in the implementation. Such gaps represent potential certification blockers, because IEC 61508 and ISO 26262 both require that every safety requirement be verified by at least one independent test or analysis. The matrix uses a three-state notation to distinguish between fully traced ( ), partially traced (◐), and not traced ( ) requirements, making gaps immediately visible to reviewers.

Third, the traceability matrix supports impact analysis when requirements change. If a new hazard is identified during operation or a safety requirement is modified due to a change in the operational environment, the matrix enables rapid identification of all downstream artefacts that are affected: which invariants must be reviewed, which components must be reassessed, which verification properties must be re-run, and which test cases must be updated. This capability is essential for maintaining the safety case over the full product lifecycle, as required by IEC 61508-1 §7.7 (modification procedures) and ISO 26262-8 §8 (change management). Without bidirectional traceability, any change to the hazard analysis would require a complete manual review of the entire specification suite, which is both error-prone and prohibitively expensive.

The scope of this matrix covers the full AMAGI Target of Evaluation (TOE) as defined in the Framework Specification v2.0. The matrix traces from HARA hazards H-01 through H-08 through six invariants (INV-001 through INV-006) to verification evidence documented in the NIPU Specification v1.1 and MEM Specification v1.1. The matrix does not trace implementation-level requirements (e.g., RTL coding standards, timing constraints, or physical design rules), which are outside the scope of the architectural traceability required by IEC 61508 and ISO 26262 at the system level.

Section 3: Traceability Methodology

The AMAGI traceability matrix employs a bidirectional traceability approach that is required by IEC 61508-1 §7.9 and recommended by ISO 26262-4 §7.4.1. Bidirectional traceability means that every link in the safety argument chain can be traced in both directions: forward from hazards to verification evidence, and backward from verification evidence to hazards. This dual-direction approach ensures both completeness and necessity of the safety argument, as described in the following subsections.

3.1 Forward Traceability (Completeness)

Forward traceability traces the path from each identified hazard through safety requirements, invariants, hardware components, and formal verification properties. The forward chain answers the question: for every hazard that has been identified, is there a complete chain of evidence demonstrating that the hazard is mitigated? The forward traceability path is: Hazard → Safety Requirement → Invariant → Component → Verification Property → Verification Method. Each link in this chain must be explicitly documented and justified. A break in any link constitutes a gap in the safety argument that must be addressed before certification can proceed. Forward traceability ensures completeness: every hazard has at least one mitigation path, and no hazard is left unaddressed.

The forward traceability chain also serves a secondary purpose: it enables prioritization of verification activities. If a hazard has only a single forward traceability path (a single invariant that mitigates it), then the verification of that invariant is critical and must be prioritized. Conversely, if a hazard has multiple forward traceability paths (multiple invariants mitigate it), then there is defence-in-depth, and the failure of any single verification property does not immediately compromise the safety argument for that hazard. The AMAGI architecture is designed to provide defence-in-depth for the most critical hazards (H-01 through H-03), where multiple invariants provide overlapping mitigation coverage.

3.2 Backward Traceability (No Gold-Plating)

Backward traceability traces the path from each verification property back through components, invariants, safety requirements, and hazards. The backward chain answers the question: for every verification activity that has been performed, is there a clear justification in terms of a safety requirement and an identified hazard? The backward traceability path is: Verification Method → Verification Property → Component → Invariant → Safety Requirement → Hazard. Backward traceability ensures necessity: no verification activity exists without a clear link to a safety requirement, preventing gold-plating (the addition of unnecessary verification activities that increase cost without improving safety).

Backward traceability is particularly important in the AMAGI context because the architecture uses formal verification methods (SPIN model checking) that are expensive to develop and maintain. If a formal verification property cannot be traced backward to a safety requirement, it may represent unnecessary effort that could be redirected to higher-priority verification activities. The backward traceability analysis in this matrix confirms that all 30 LTL properties in the SPIN models can be traced backward to at least one safety requirement, ensuring that no formal verification effort is wasted on properties that do not contribute to the safety argument.

3.3 Traceability Status Notation

The traceability matrix uses a three-state notation to indicate the status of each traceability link. This notation provides an at-a-glance summary of the completeness of the safety argument and makes gaps immediately visible to reviewers and auditors. The three states are defined as follows:

Symbol	Status	Meaning	Certification Implication
	Fully Traced	The safety requirement has a complete forward traceability chain from hazard through invariant, component, and verification property. The verification property has been formally proven or empirically validated.	
◖	**Partially Traced**	The safety requirement has a forward traceability chain, but verification evidence exists only as design analysis or environmental testing without formal proof. Additional verification is needed for certification.	
	Not Traced	The safety requirement has no verification evidence, or the existing evidence is insufficient to confirm that the invariant holds. This is a gap that must be remediated before certification.	

The use of this three-state notation is consistent with the approach recommended by the ISO 26262-4 §7.4.1 requirement for bidirectional traceability, and it aligns with the IEC 61508-3 §7.4.2 requirement for verification of safety requirements. A fully traced link ( ) satisfies both standards. A partially traced link (◖) indicates that the requirement is addressed but not at the rigour level required for the target SIL or ASIL. A not-traced link ( ) indicates a gap that must be remediated before the safety case can be considered complete.

Section 4: Master Traceability Matrix

The Master Traceability Matrix is the central artefact of this document. It provides a complete mapping from each hazard identified in the HARA through safety requirements, invariants, hardware components, specification sections, verification properties, and verification methods to a traceability status indicator. The matrix is designed to be read both horizontally (following a single hazard through its complete traceability chain) and vertically (identifying all hazards and safety requirements that depend on a particular invariant or component). Each row represents one safety requirement and its complete traceability chain; where a hazard maps to multiple safety requirements, each requirement occupies its own row, ensuring that no traceability link is ambiguous.

The matrix uses a compact notation to fit within the page width. Specification sections are referenced by document abbreviation and section number (e.g., NIPU §3.3 refers to Section 3.3 of the NIPU Specification v1.1). Verification properties are identified by their SPIN model designators (e.g., SCC-S01 through SCC-S06) or by descriptive names for non-SPIN verification methods. The Status column uses the three-state notation defined in Section 3.3. The table below contains the complete master traceability matrix for all eight hazards and sixteen safety requirements in the AMAGI safety argument.

Hazard	Safety Req	Invariant(s)	Component	Spec Section	Verification Property	Verification Method	Status
H-01	SR-01: TTP non-bypassable	INV-001, INV-005	SCC + EGW	NIPU §3.3, §3.5	SCC-S01 through SCC-S06	SPIN model (25.9M states)	

Hazard	Safety Req	Invariant(s)	Component	Spec Section	Verification Property	Verification Method	Status
H-01	SR-02: Dual approval for actuation	INV-003, INV-005	EGW	NIPU §3.3	EGW AND-gate truth table	Online hardware test	
H-02	SR-03: MAG immutability	INV-004	MAG OTP	MEM §3.2	CRTM Golden Hash comparison	Boot-time verification	
H-02	SR-04: Post-silicon MAG validation	INV-004, INV-005	CRTM	NIPU §3.3	CRTM hash match	Boot test (ADR-007)	
H-03	SR-05: All memory access mediated	INV-005	MPE + ATL	MEM §3.1, §3.3	MPE-S01 through MPE-S06	SPIN model	
H-03	SR-06: Secure/Normal World separation	INV-003	MPE, MAG	MEM §3.2	MPE-S03 (Source ID internal)	SPIN model	
H-04	SR-07: Audit trail integrity	INV-006	ATL + Amagi-Q	MEM §3.3, §3.5	ATL-S01 through ATL-S05	SPIN model	
H-04	SR-08: Audit diode unidirectionality	INV-006	ATL audit diode	MEM §3.3	TX-only hardware gateway	Design review	
H-05	SR-09: DDL delay non-shortening	INV-001, INV-002	PVT-calibrated DDL	NIPU §3.5	DDL delay bound checking	Fault injection (IEC 61508-2 Annex A)	◖
H-05	SR-10: PVT stability of delay	INV-005	DDL replica-path	NIPU §3.5	PVT calibration feedback	Environmental testing	◖
H-06	SR-11: Kill switch activation within window	INV-001, INV-005	Termination latch + AFC	NIPU §3.4	12–35 μs cutoff measurement	Hardware timing test	
H-06	SR-12: AFC independent fallback	INV-001	AFC (analog)	NIPU §3.4	AFC heartbeat watchdog	Fault injection	
H-07	SR-13: Supply chain Trojan detection	INV-004, INV-005	CRTM	NIPU §3.3	Golden Hash comparison	Boot-time verification	
H-07	SR-14: Netlist-to-RTL equivalence	INV-002	SCC, MPE	S-APL §13.1	Post-synthesis equivalence	Formal equivalence check	
H-08	SR-15: Rollback prevention	INV-004	Amagi-Q OTA	NIPU §3.6	Monotonic counter	Formal verification	◖
H-08	SR-16: PQC-signed policy updates	INV-004, INV-006	Amagi-Q	MEM §3.5	ML-DSA-44 signature verification	Known-answer test	

Summary: Of the 16 safety requirements traced in this matrix, 11 are fully traced ( ), 3 are partially traced (◖), and 2 are not traced ( ). The partially traced requirements are SR-09 (DDL delay non-shortening), SR-10 (PVT stability of delay), and SR-15 (rollback prevention). The not-traced requirements are SR-08 (audit diode unidirectionality) and SR-14 (Netlist-to-RTL equivalence). The four gaps are addressed in the Gap Remediation Plan (Section 6).

Section 5: Coverage Analysis

This section presents a summary analysis of the traceability coverage organized by invariant. The coverage analysis reveals how each invariant contributes to the overall safety argument, which hazards and safety requirements it addresses, and how many gaps remain in its verification evidence. This invariant-centric view complements the hazard-centric view of the Master Traceability Matrix (Section 4) by highlighting the critical invariants that provide the most coverage and the invariants that have the most gaps requiring remediation. Understanding the coverage distribution is essential for prioritizing verification efforts, because an invariant that addresses multiple hazards and has zero gaps is a strong point in the safety argument, while an invariant with gaps may represent a single point of failure.

The table below summarizes the invariant coverage. The “Hazards Covered” column lists all hazards for which the invariant provides at least partial mitigation. The “Safety Requirements” column lists the safety requirements that are directly implemented by the invariant. The “Verification Properties” column identifies the specific SPIN properties, hardware tests, or design analyses that verify the invariant. The “Gap Count” column indicates the number of safety requirements for which the invariant contributes to the verification but where the verification evidence is incomplete (partial or missing). A gap count of zero means the invariant is fully verified for all safety requirements it addresses.

Invariant	Hazards Covered	Safety Requirements	Verification Properties	Gap Count
INV-001	H-01, H-02, H-03, H-04, H-05, H-06	SR-01, SR-09, SR-11, SR-12	SCC-S01–S06, DDL check, cutoff test	0
INV-002	H-05, H-07	SR-09, SR-14	DDL bound, equivalence check	**1** (SR-14 not yet verified)
INV-003	H-01, H-03	SR-02, SR-06	EGW test, MPE-S03	0
INV-004	H-02, H-07, H-08	SR-03, SR-04, SR-13, SR-15, SR-16	CRTM hash, counter monotonicity	**1** (SR-15 partial)
INV-005	H-01, H-02, H-03, H-05, H-06	SR-01, SR-05, SR-10	MPE-S01–S06, DDL PVT	**1** (SR-10 partial)
INV-006	H-04, H-08	SR-07, SR-08, SR-16	ATL-S01–S05, audit diode	**1** (SR-08 not verified)

The coverage analysis identifies a total of four gaps across the six invariants. Two invariants (INV-001 and INV-003) have zero gaps and represent the strongest points in the safety argument. Three invariants (INV-002, INV-004, and INV-005) each have one gap, and one invariant (INV-006) has one gap. The specific gaps are: SR-08 (audit diode unidirectionality) in INV-006, where no verification evidence exists beyond a design review; SR-14 (Netlist-to-RTL equivalence) in INV-002, where formal equivalence checking has not yet been performed; SR-10 (PVT environmental testing) in INV-005, where environmental testing is planned but not yet completed; and SR-15 (counter monotonicity formal proof) in INV-004, where the SPIN model has not yet been extended to include OTA counter behaviour. All four gaps are addressed in the Gap Remediation Plan (Section 6) with specific actions, timelines, and responsible parties.

It is noteworthy that INV-001 (Physical Non-Bypassability) and INV-003 (HW Separation of Control and Cognition) have zero gaps despite covering multiple hazards and safety requirements. This is because the SPIN model checking of the SCC FSM and the MPE+ATL combined model provides comprehensive formal verification of these invariants. INV-005 (Complete Mediation) is the most broadly applicable invariant, covering five of the eight hazards, and its single gap (SR-10) is a testing gap rather than a design gap, meaning the architectural mechanism exists but has not yet been tested under the full range of environmental conditions.

Section 6: Gap Remediation Plan

This section provides a specific remediation action for each of the four gaps identified in the coverage analysis. Each remediation action includes the gap identifier, the safety requirement that is not fully verified, the nature of the gap, the planned remediation activity, the target milestone for completion, and the justification for why the remediation will close the gap. The remediation plan is prioritized by the severity of the gap: gaps in safety requirements that address the most critical hazards (H-01 through H-04) are remediated first, followed by gaps in requirements that address lower-criticality hazards (H-05 through H-08).

6.1 Gap: SR-08 — Audit Diode Unidirectionality

Safety requirement SR-08 mandates that the ATL audit diode be unidirectional, ensuring that no data can flow from the audit trail back into the operational domain. The current verification evidence consists of a design review that confirms the architectural intent of the diode, but does not provide physical evidence that the diode is implemented correctly and cannot be subverted. This gap is particularly important because the audit diode is the primary mechanism for enforcing INV-006 (Complete Audit Integrity), and if the diode can be reversed, an adversarial AI workload could potentially tamper with audit records, undermining the entire audit trail architecture.

Remediation Action: Electrical testing of the diode under reverse-bias and forward-bias conditions to confirm unidirectional current flow. The reverse-bias voltage test verifies that no leakage current exceeds the specification threshold, while the forward-bias test confirms nominal conduction. This testing is scheduled for the pre-prototype phase, prior to first silicon. Additionally, a test chip with a dedicated diode test structure will be included in the first prototype to enable full electrical characterization of the diode under all operating conditions, including temperature extremes and supply voltage variation.

6.2 Gap: SR-14 — Netlist-to-RTL Equivalence

Safety requirement SR-14 requires that the synthesized netlist be formally equivalent to the RTL source, ensuring that the synthesis process has not introduced any functional changes. This requirement is mandated by S-APL §13.1 and is a standard requirement in safety-critical hardware development under IEC 61508-2 and DO-254. The current gap is that no formal equivalence checking has been performed, because the RTL implementation has not yet been completed. This is expected at the specification stage, but the remediation plan must ensure that equivalence checking is performed as soon as the RTL becomes available.

Remediation Action: Formal equivalence checking using an industry-standard tool (e.g., Cadence Conformal, Synopsys Formality) per S-APL §13.1. The equivalence check will be performed as a mandatory gate in the design flow, and no design will be allowed to proceed to physical implementation without a passing equivalence check result. This remediation is required before Tier 1 certification and is a hard dependency for the certification timeline. The equivalence checking scripts and results will be archived as certification evidence under ALC_CMS.2 (Configuration Management) and ADV_TDS.1 (Design Documentation).

6.3 Gap: SR-10 — PVT Environmental Testing

Safety requirement SR-10 requires that the DDL (Deterministic Delay Line) delay remain stable across the full range of Process, Voltage, and Temperature (PVT) conditions. The current verification evidence consists of architectural analysis and PVT calibration feedback design, but no physical testing under environmental extremes has been performed. This gap exists because the DDL is a mixed-signal circuit whose behaviour cannot be fully verified by formal methods alone; it requires physical measurement under controlled environmental conditions. The PVT calibration mechanism is designed to compensate for environmental variation, but the effectiveness of this compensation must be empirically validated.

Remediation Action: Full temperature cycle testing from −40°C to +125°C with DDL delay measurement at each temperature point, performed on the FPGA prototype. The test will measure the DDL delay at a minimum of 10 temperature points across the full range, with supply voltage variation of ±10% at each temperature point. The measured delay values will be compared against the specification bounds defined in the NIPU Specification v1.1 §3.5. This testing is planned for the FPGA prototype phase and will be documented in the AMAGI Verification Report as evidence for SR-10 compliance.

6.4 Gap: SR-15 — Counter Monotonicity Formal Proof

Safety requirement SR-15 requires that the OTA (Over-The-Air) update counter be strictly monotonic, preventing rollback attacks that could revert the device to a previously compromised firmware version. The current verification evidence consists of a design analysis that confirms the architectural intent of the monotonic counter, but no formal proof exists that the counter implementation is correct. This gap is important because a rollback attack could undermine all other security mechanisms by reverting the firmware to a version with known vulnerabilities. The counter monotonicity must be verified at the same level of rigour as the other AMAGI invariants.

Remediation Action: Extend the SPIN model to include OTA counter behaviour, creating a formal model of the counter increment, storage, and verification operations. The model will include an LTL property asserting that the counter value never decreases across any sequence of operations, including power cycles and fault injection scenarios. This work is planned for the v1.2 verification package and will be integrated with the existing SPIN verification suite. The extended model will also verify that the ML-DSA-44 signature verification correctly validates the counter value as part of the OTA update process, providing end-to-end formal verification of the rollback prevention mechanism.

6.5 Gap Remediation Summary

Gap	Safety Req	Current Status	Remediation Action	Target Milestone
1	**SR-08**	Not traced (design review only)	Electrical test: reverse-bias and forward-bias voltage measurement of diode; test chip with diode test structure	Pre-prototype phase
2	**SR-14**	Not traced (RTL not yet available)	Formal equivalence checking (Conformal/Formality) per S-APL §13.1	Before Tier 1 certification
3	**SR-10**	◖ Partially traced (analysis only)	Full temperature cycle testing (−40°C to +125°C) with DDL delay measurement	FPGA prototype phase
4	**SR-15**	◖ Partially traced (design analysis)	Extend SPIN model to include OTA counter behaviour with LTL monotonicity property	v1.2 verification package

Section 7: Cross-Document References

The AMAGI specification suite consists of multiple documents that collectively define the architecture, its implementation requirements, its verification evidence, and its regulatory alignment. Each document contributes to the traceability matrix by providing specific sections that define hazards, safety requirements, invariants, components, or verification properties. This section maps each AMAGI document to the traceability entries it supports, enabling reviewers to quickly locate the relevant specification text for any traceability link in the Master Traceability Matrix.

The cross-document reference table below lists all documents in the AMAGI specification suite, their current versions, the invariants they address, and the safety requirements they trace. Documents that provide cross-reference-only support (e.g., regulatory mapping documents) are included for completeness but do not contain direct traceability links to specific invariants or safety requirements. The Master Parameter Sheet v2.0 is listed separately because it provides quantitative parameter values that support the verification of all safety requirements, even though it does not define any requirements or invariants directly.

Document	Version	Relevant Invariants	Relevant Safety Requirements
Framework Specification	v2.0	All (INV-001 through INV-006)	All (SR-01 through SR-16)
NIPU Specification	v1.1	INV-001, INV-002, INV-003, INV-005	SR-01, SR-02, SR-09, SR-10, SR-11, SR-12
MEM Specification	v1.1	INV-001, INV-004, INV-005, INV-006	SR-03, SR-05, SR-06, SR-07, SR-08
HARA	v1.0	All	All
FMEDA	v1.0	INV-001, INV-005, INV-006	SR-01, SR-09, SR-11
S-APL	v2.0	INV-002, INV-004	SR-14, SR-03
NIST AI RMF Profile	v1.1	Regulatory mapping	Cross-reference only
Charter	v1.1	Governance	Cross-reference only

Document	Version	Relevant Invariants	Relevant Safety Requirements
Open Specification	v1.1	Interoperability	Cross-reference only
IAA	v1.1	Class definition	Cross-reference only
Master Parameter Sheet	v2.0	All (quantitative); DOI 10.5281/zenodo.20145212	All (parameter verification)

The cross-document reference table confirms that the Framework Specification v2.0 provides the broadest coverage, addressing all six invariants and all sixteen safety requirements. The NIPU and MEM specifications together cover the full set of invariants and safety requirements, with the NIPU specification focusing on processor-level mechanisms (SCC, EGW, DDL, AFC, CRTM) and the MEM specification focusing on memory-level mechanisms (MAG, MPE, ATL, Amagi-Q). The HARA and FMEDA provide orthogonal evidence: the HARA defines the hazards that initiate the traceability chain, and the FMEDA provides quantitative evidence that the safety mechanisms achieve the required Safety Integrity Level. The S-APL, NIST AI RMF Profile, Charter, Open Specification, and IAA provide supporting context but do not contain direct traceability links to specific invariants or safety requirements.

Section 8: Maintenance and Update Procedure

The traceability matrix is a living document that must be maintained throughout the AMAGI product lifecycle. IEC 61508-1 §7.7 requires that the safety case (of which the traceability matrix is a key component) be updated whenever there is a change to the hazards, safety requirements, or safety-related systems. ISO 26262-8 §8 requires a change management process that includes impact analysis on all affected safety artefacts. The following procedure defines when and how the traceability matrix shall be updated to maintain its integrity and currency.

The traceability matrix shall be updated under the following four conditions. First, whenever a new hazard is identified in the HARA, the matrix must be extended with new rows that trace the hazard through safety requirements, invariants, components, and verification properties. The Lead Auditor shall verify that the new hazard does not create any orphan requirements (safety requirements without invariants) or orphan verification properties (properties without safety requirements). Second, whenever a new safety requirement is derived from an existing or new hazard, the matrix must be updated with a new row that traces the requirement through its complete chain. The Lead Auditor shall verify that the new requirement maps to at least one invariant and that at least one verification property exists for the invariant.

Third, whenever a verification property is added, modified, or removed in the SPIN model or any other verification environment, the matrix must be updated to reflect the change. If a verification property is removed, the Lead Auditor shall verify that the affected safety requirements still have at least one other verification property, or shall record a gap in the matrix. If a verification property is modified, the Lead Auditor shall verify that the modified property still provides adequate evidence for the safety requirements it traces to. Fourth, at each version increment of any AMAGI specification document (Framework, NIPU, MEM, HARA, FMEDA, S-APL), the matrix shall be reviewed to determine whether any changes to the document affect the traceability links. If changes are identified, the matrix shall be updated and the version of this document shall be incremented accordingly.

The AGIAM Charter §5.2 designates the Lead Auditor as the person responsible for maintaining traceability integrity. The Lead Auditor shall perform a traceability review at each major project milestone (specification review, design review, prototype review, certification readiness review) and shall report any gaps or inconsistencies to the AGIAM governance board. All updates to the traceability matrix shall be recorded in a change log that identifies the date, the reason for the update, the specific rows or columns affected, and the name of the person who made the update. The change log shall be maintained as an appendix to this document and shall be available for review by certification auditors.

In addition to event-driven updates, the traceability matrix shall be subject to an annual completeness review, even if no triggering event has occurred. This review shall verify that all rows in the matrix have a complete traceability chain, that all status indicators are current, and that all gaps identified in previous reviews have been addressed or have an active remediation plan. The annual review is a proactive measure to ensure that the matrix does not become stale due to changes in other documents that were not captured by the event-driven update procedure. The results of the annual review shall be documented in a Traceability Integrity Report and submitted to the AGIAM governance board for approval.